

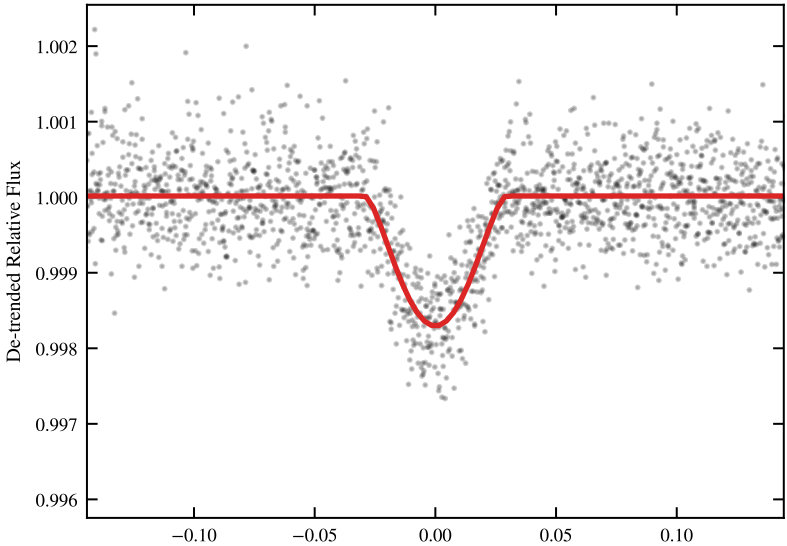
TESS PLANET TRANSIT CHARACTERIZATION REPORT

Target: TIC 154840461 | RA: 182.69156° | Dec: 85.70508° | Sectors: 19, 20

Derived Planetary Parameters

Parameter	Derived Value (MCMC)
Period (d)	6.0358 ± 0.000157
Rp/R★	0.24938 ± 0.103
Rp (R⊕)	45.886 ± 19
b	1.209 ± 0.107
T14 (h)	1.3857 ± 0.0362
a (AU)	0.083313 ± 0.00438
Teq (K)	1607.4 ± 113

Primary Phased Transit



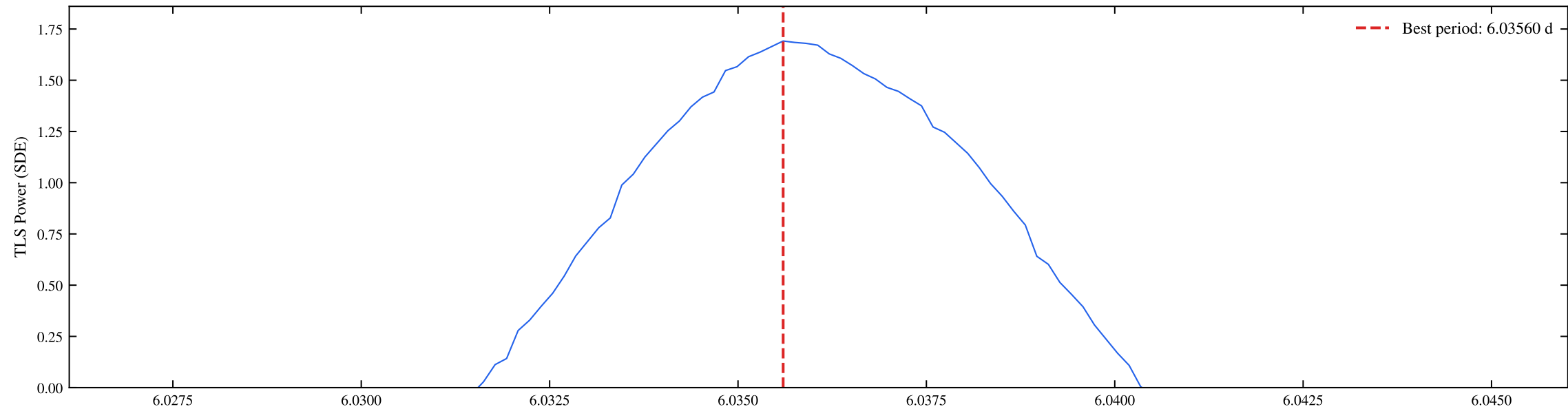
Literature Comparison & MCMC Diagnostics

Parameter / Metric	Literature Value	Derived / Diagnostic
P1 Period (d)	6.0356	6.035823 ± 0.00016
P1 Epoch (BTJD)	N/A	1818.3725 ± 0.0008
Max R-hat (R)	N/A	1.0009
Min ESS	N/A	5005
MCMC Divergences	N/A	0

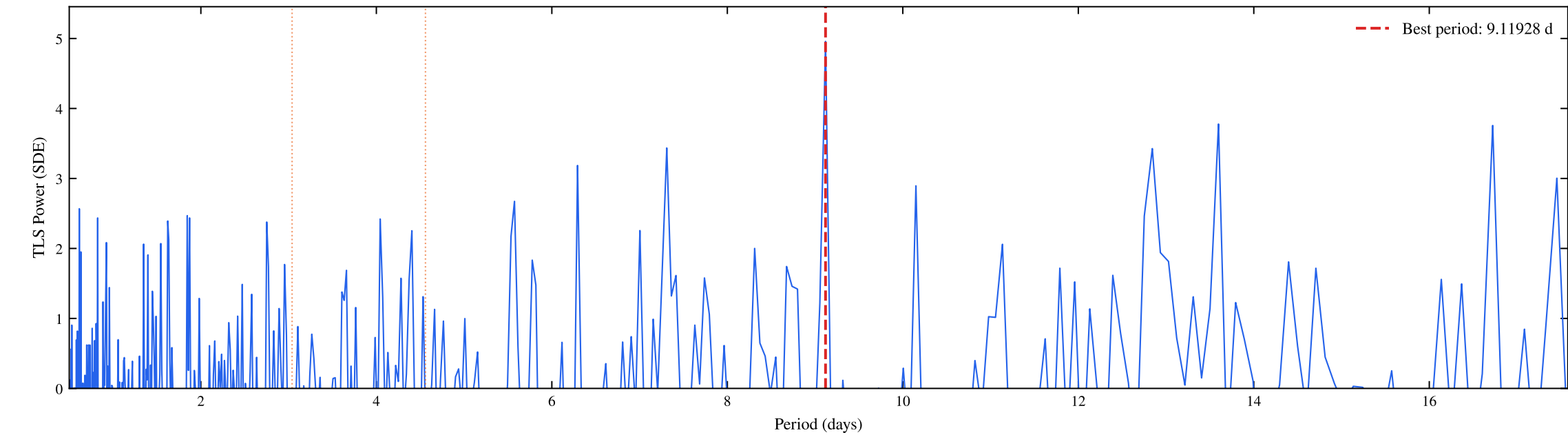
Host Star Properties

Stellar Parameter	Value	Source
R★ (R☉)	1.688 ± 0.061	VizieR TIC8
M★ (M☉)	1.689 ± 0.306	VizieR TIC8
Teff (K)	7401 ± 502	VizieR TIC8
log g (dex)	4.211 ± 0.0894	VizieR TIC8
[Fe/H]	-0.4229 ± 0.154	Gaia DR3
ρ★ (g/cm³)	0.4951 ± 0.105	Derived

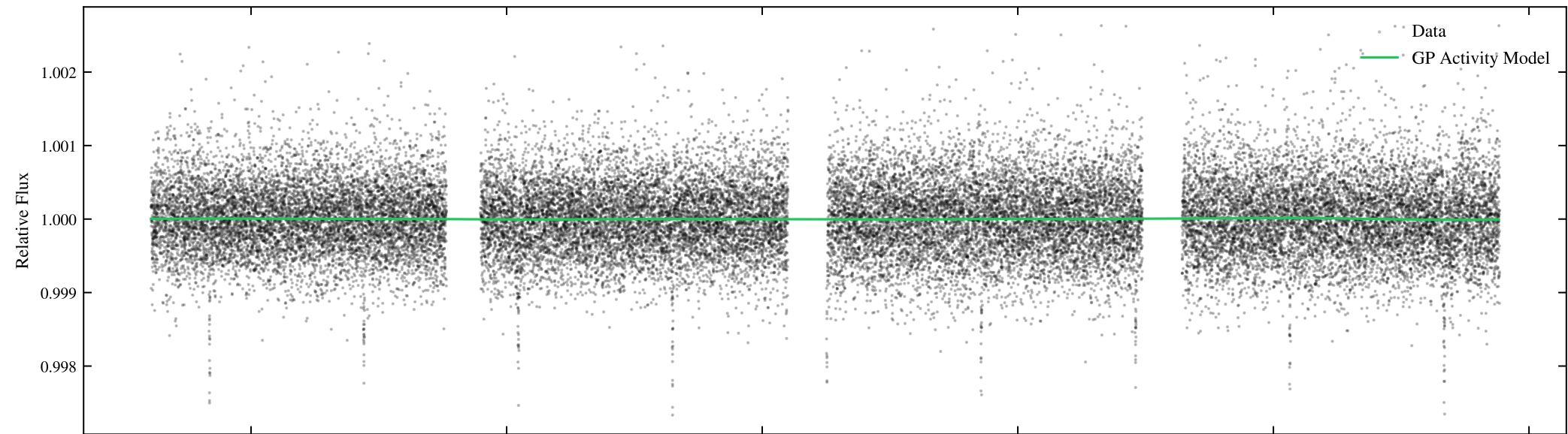
TIC 154840461 | Sectors: 19, 20 | TLS Periodogram (Planet 1 Search) (SDE = 1.69)



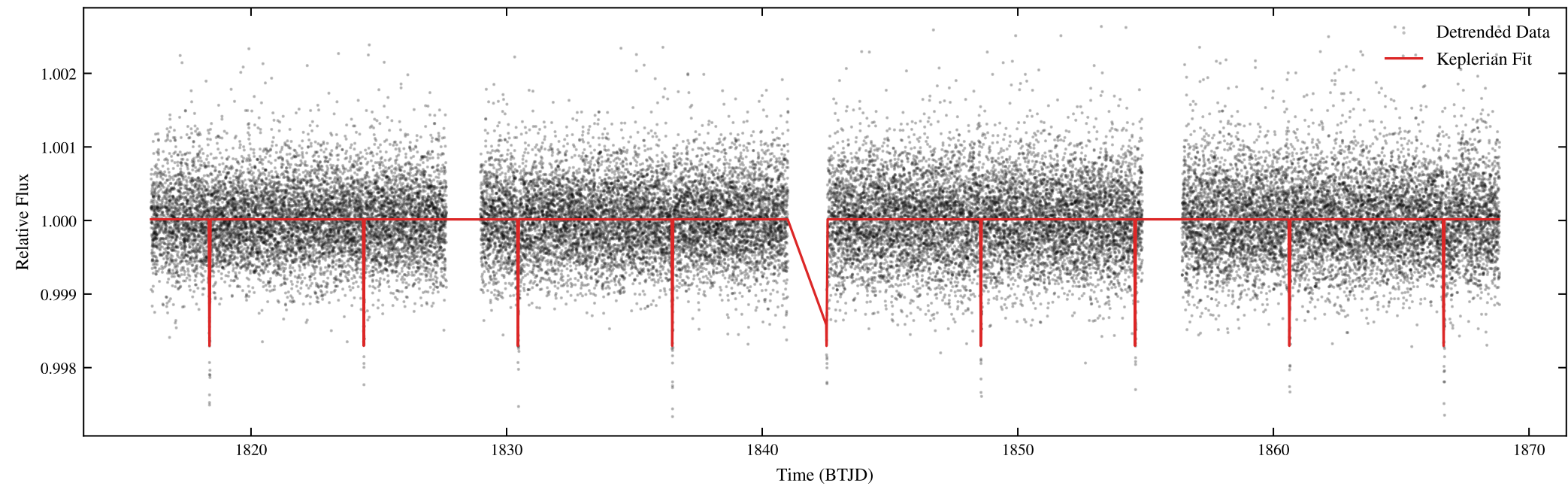
TIC 154840461 | Sectors: 19, 20 | TLS Periodogram (Planet 2 Search) (SDE = 4.96)



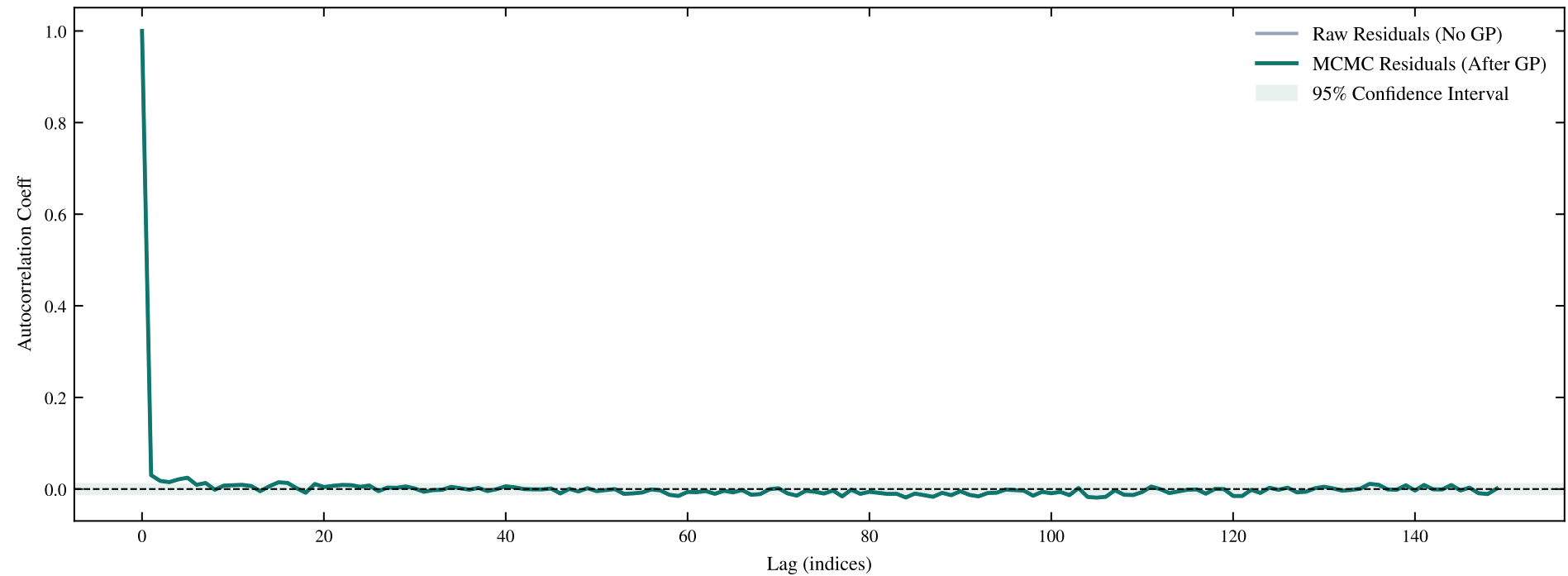
TIC 154840461 | Sectors: 19, 20 | Bayesian GP Detrending (Time Domain)



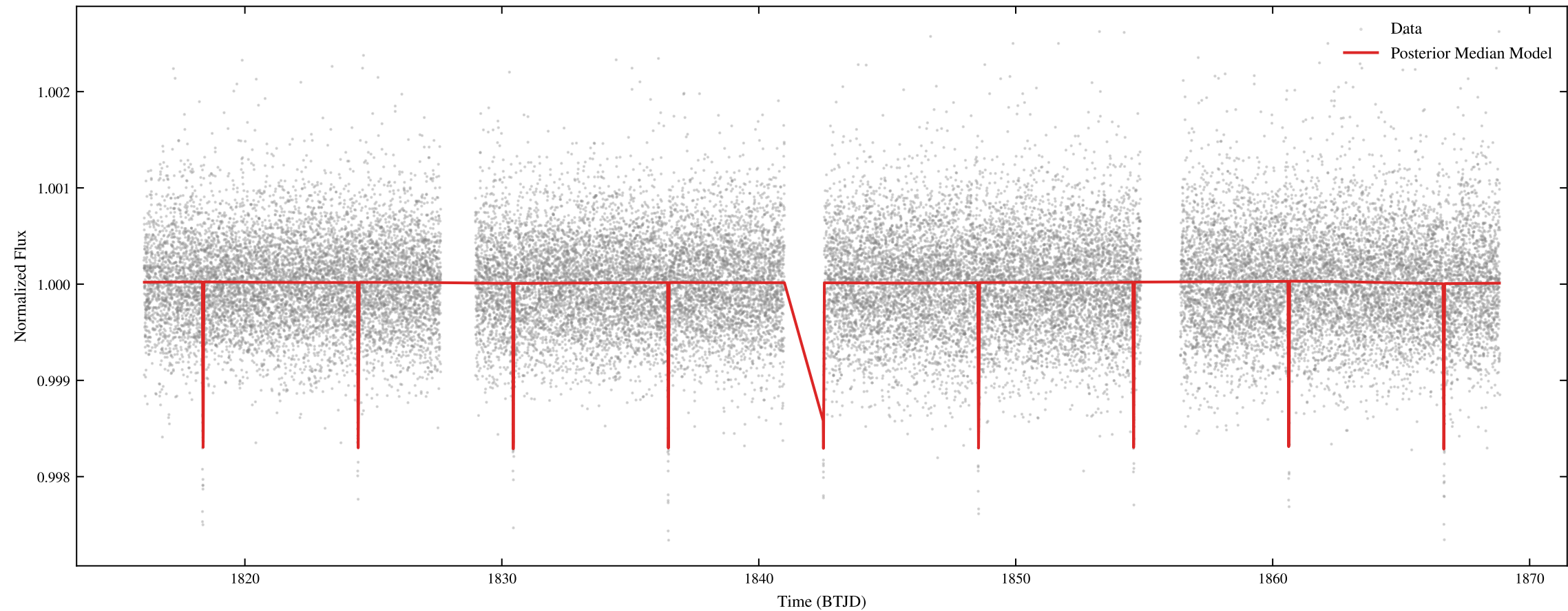
Detrended Light Curve & Best-Fit Transit Model

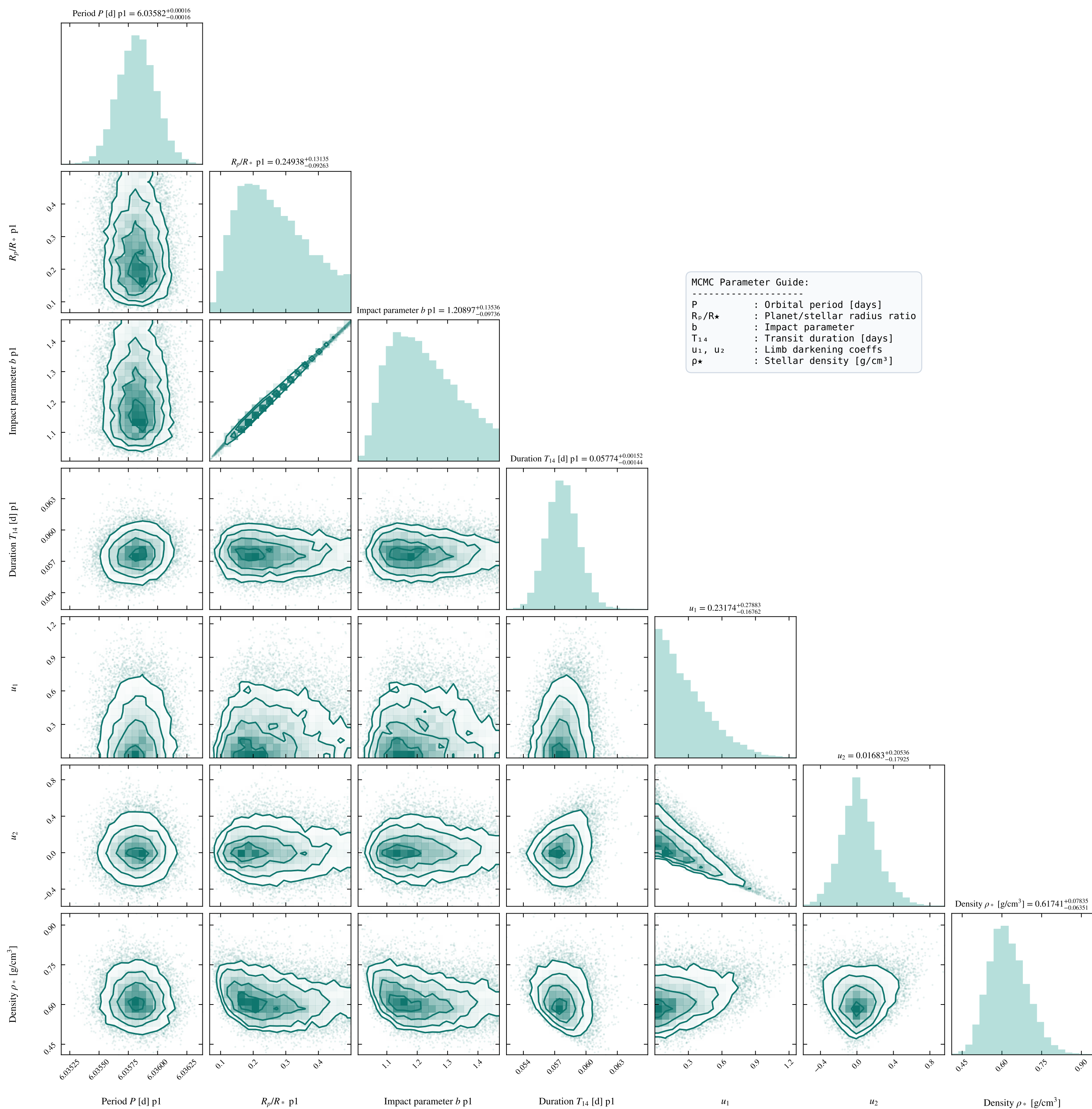


TIC 154840461 | Sectors: 19, 20 | GP Residuals Autocorrelation Function (ACF)

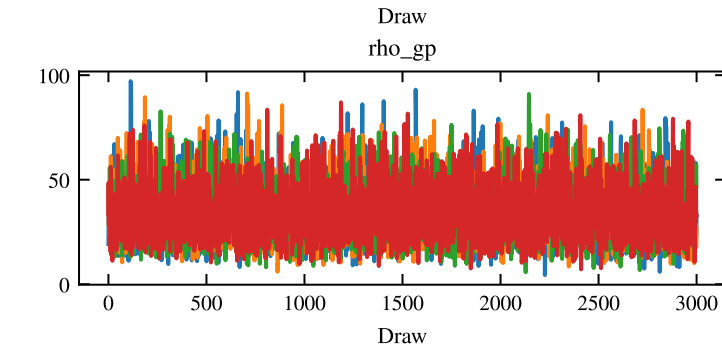
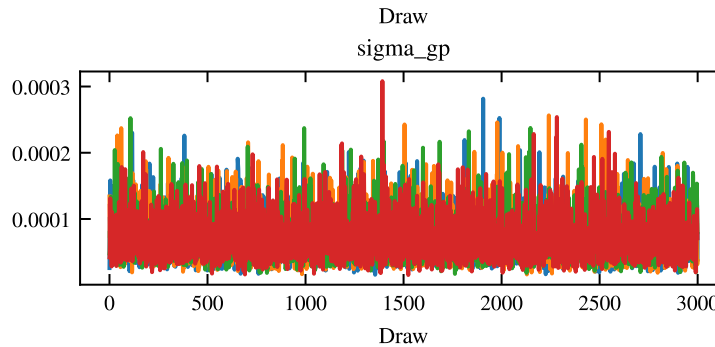
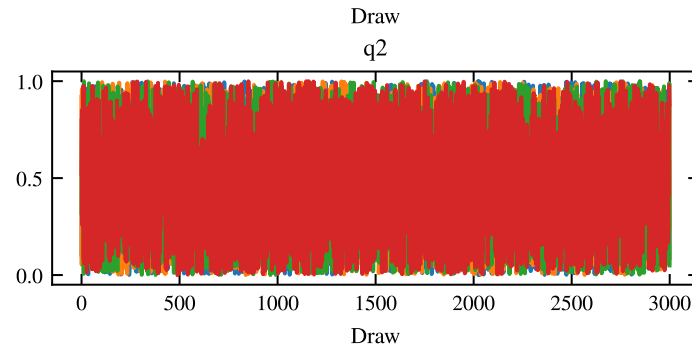
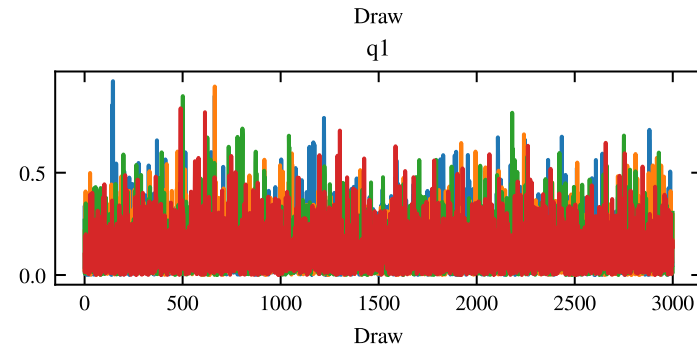
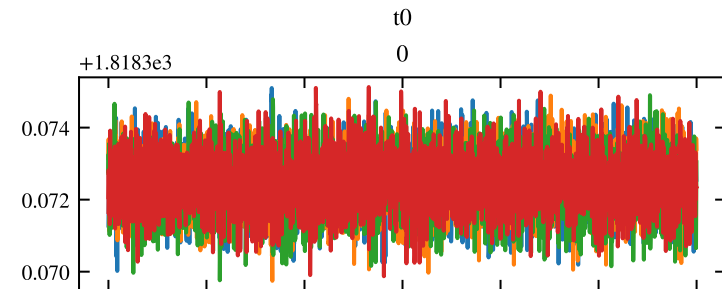
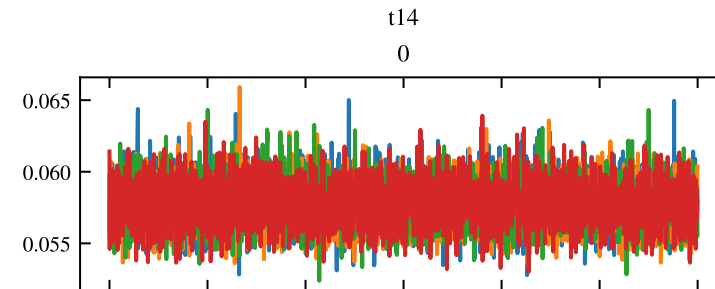
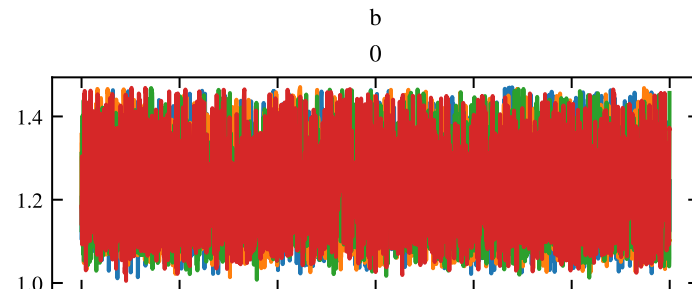
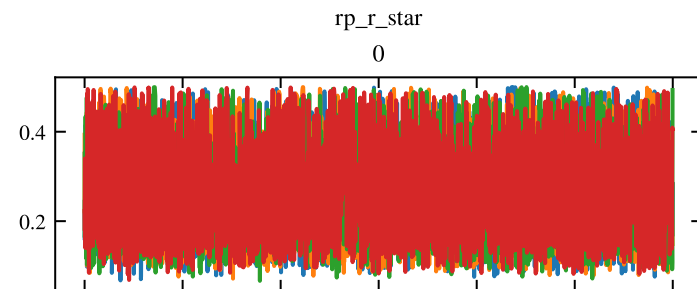


TIC 154840461 | Sectors: 19, 20 | MCMC Posterior Predictive Check

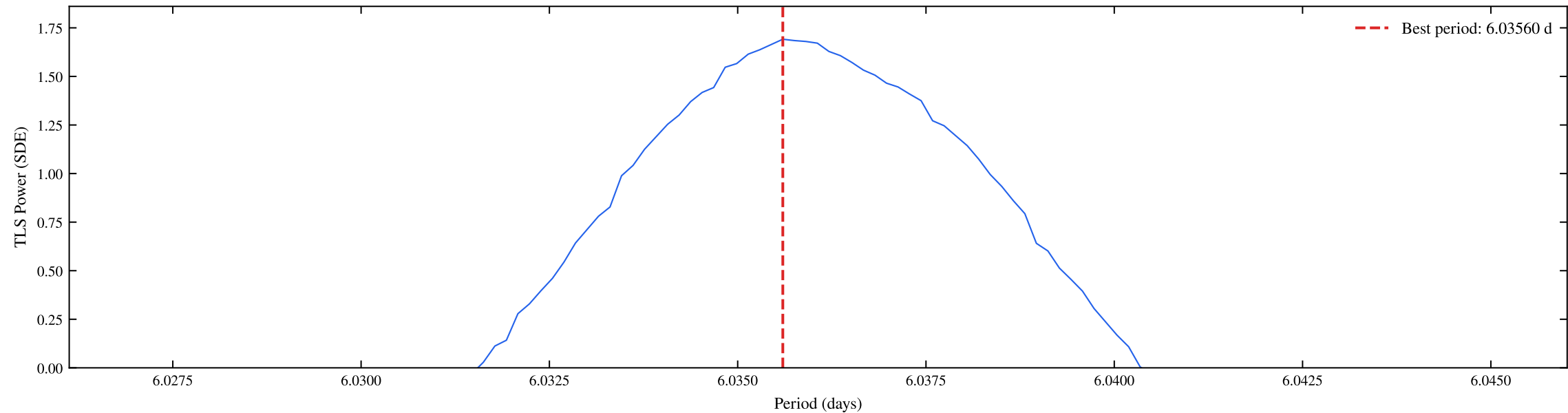




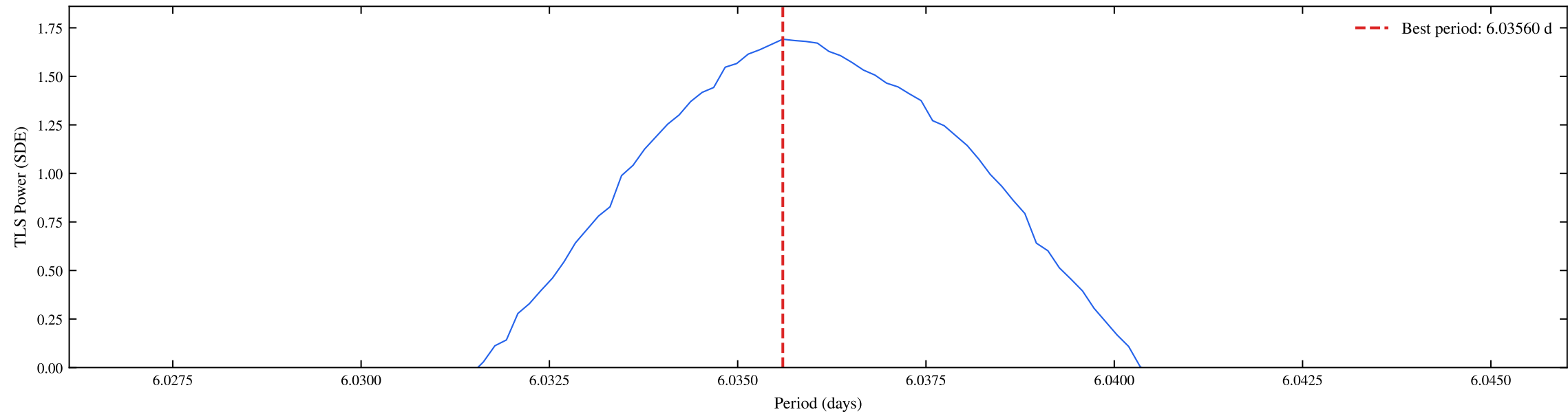
TIC 154840461 | Sectors: 19, 20 | MCMC Trace Plots



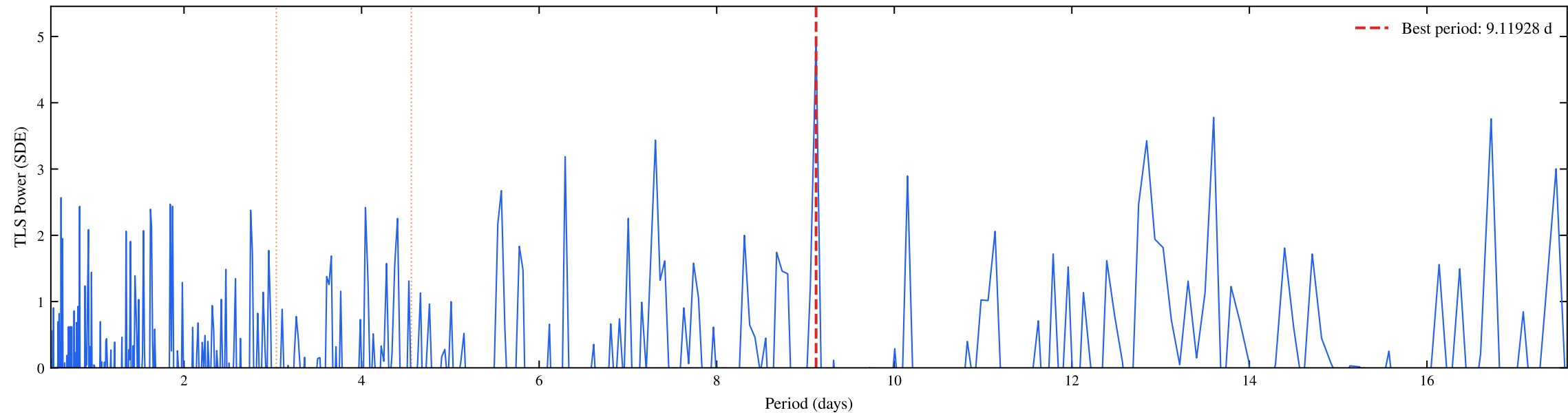
TIC 154840461 | Sectors: 19, 20 | TLS Coarse Periodogram (SDE = 1.69)



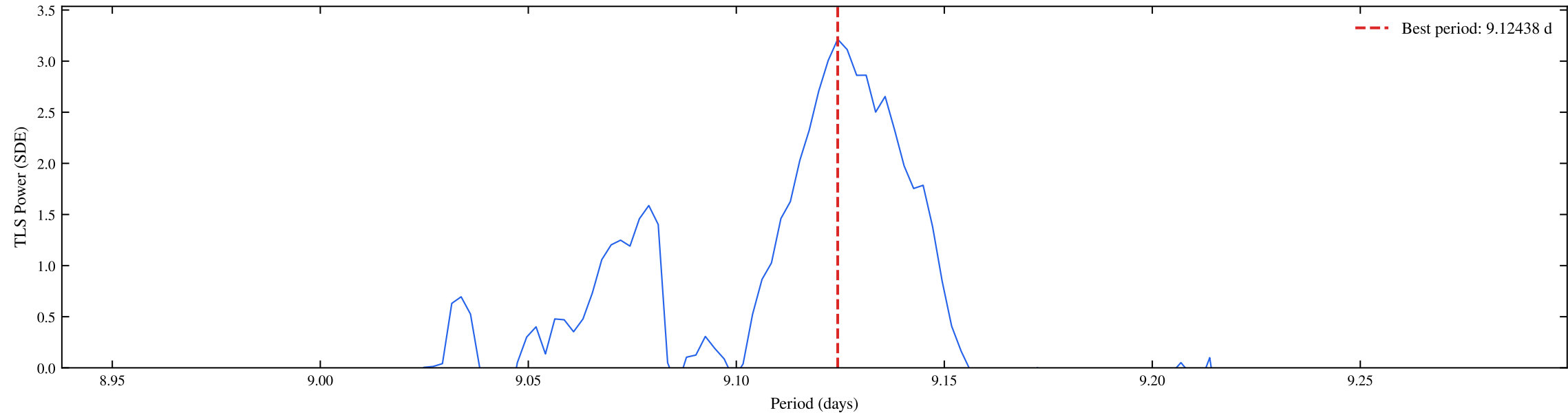
TIC 154840461 | Sectors: 19, 20 | TLS Fine Periodogram (SDE = 1.69)



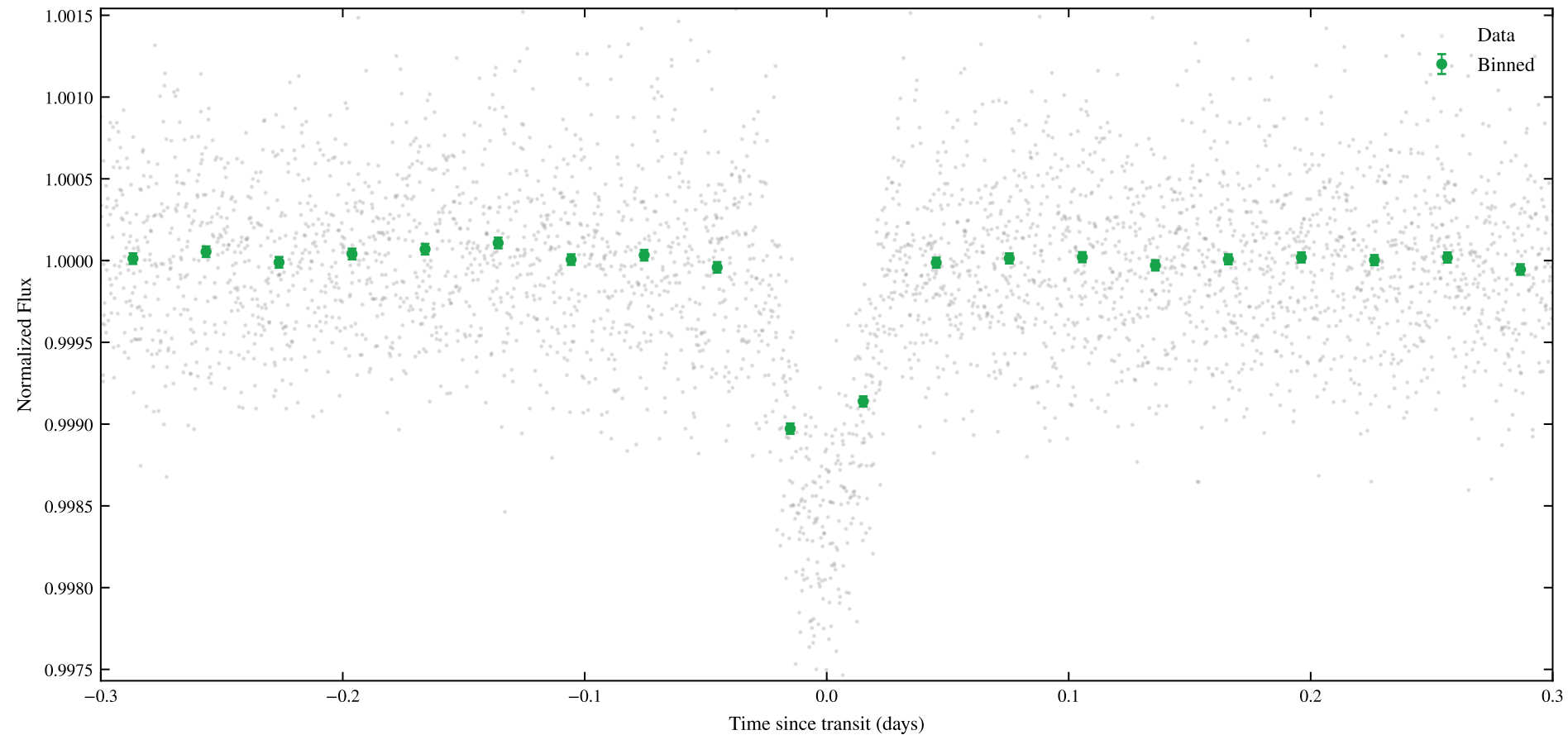
TIC 154840461 | Sectors: 19, 20 | TLS Coarse Periodogram (SDE = 4.96)



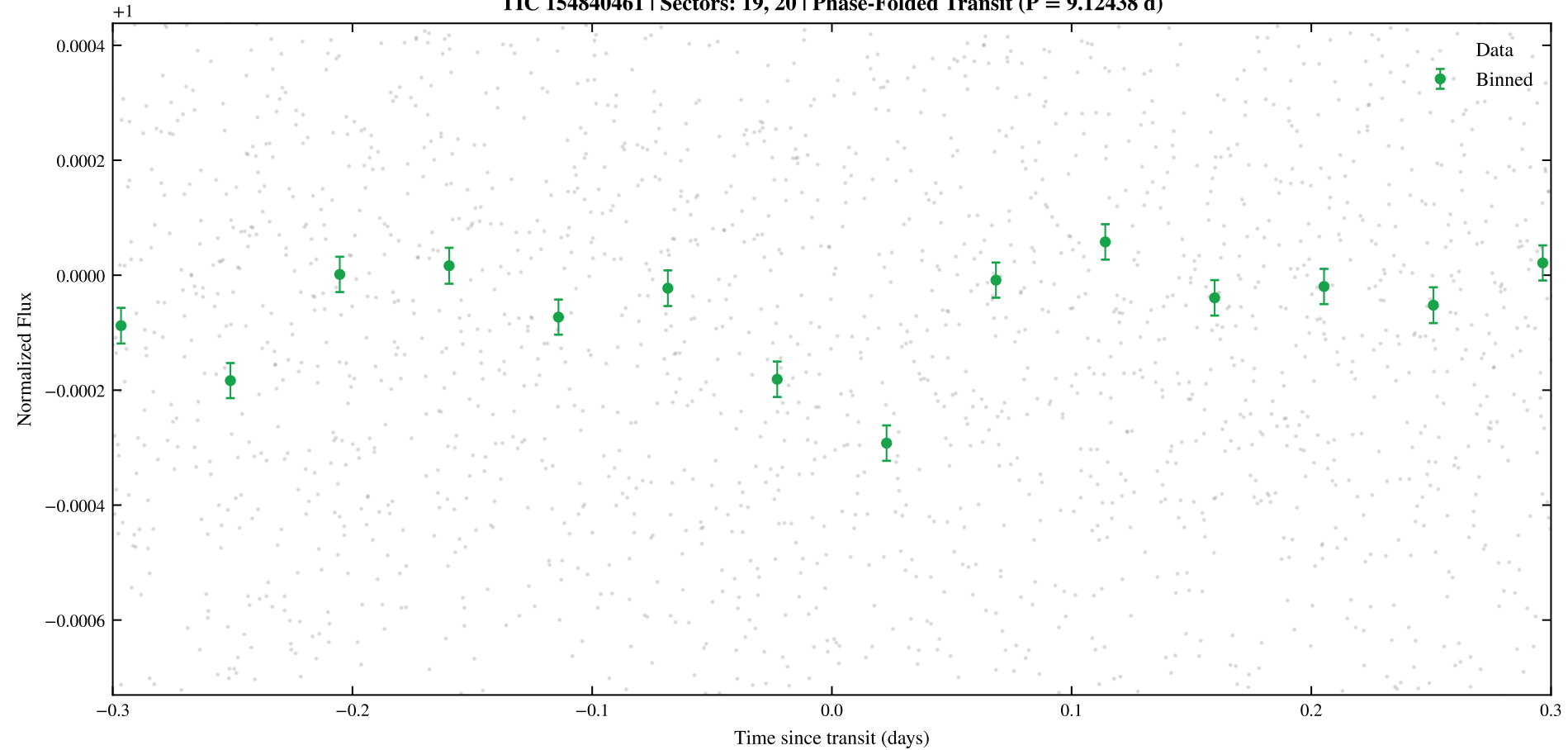
TIC 154840461 | Sectors: 19, 20 | TLS Fine Periodogram (SDE = 3.21)

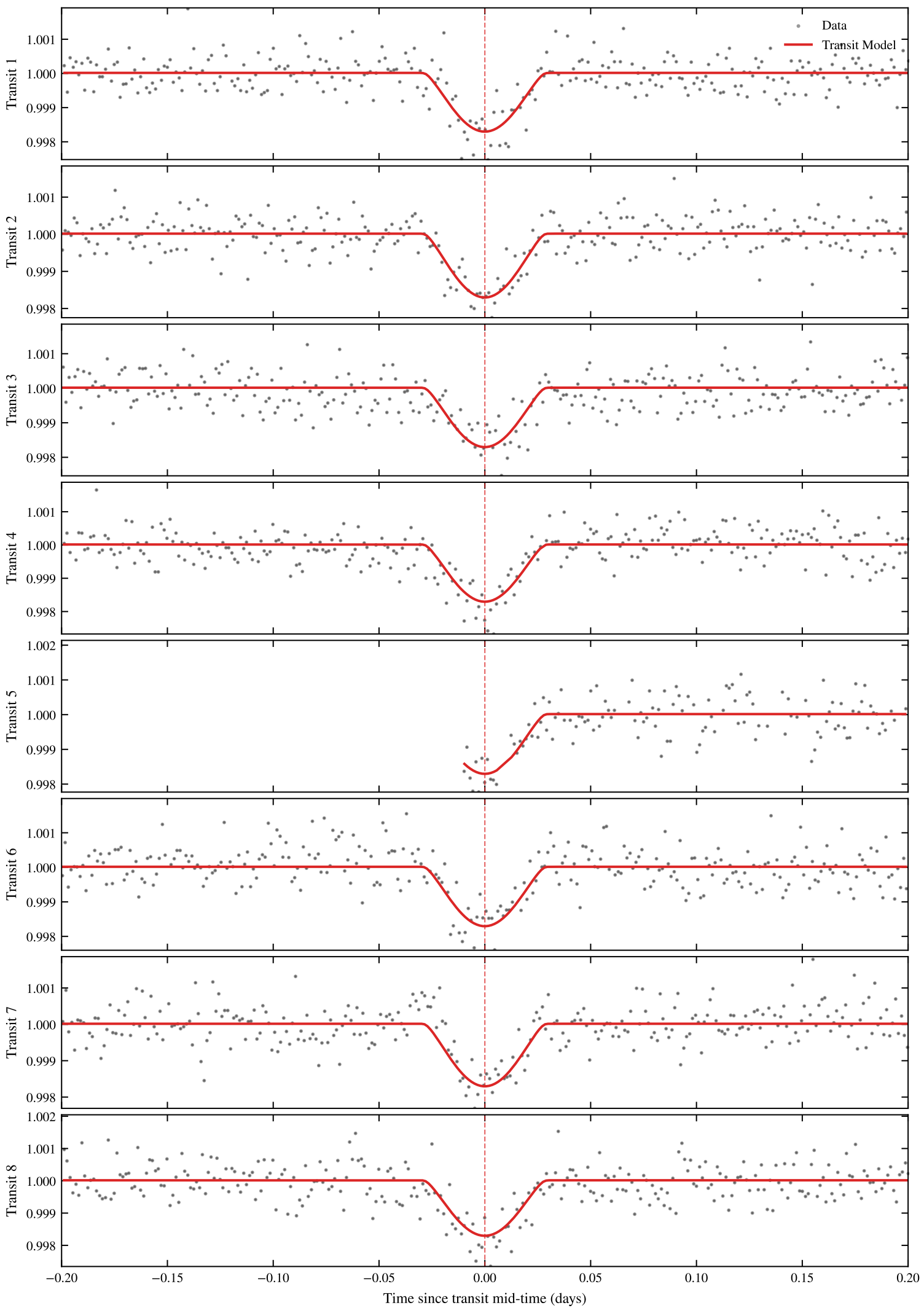


TIC 154840461 | Sectors: 19, 20 | Phase-Folded Transit (P = 6.03560 d)



TIC 154840461 | Sectors: 19, 20 | Phase-Folded Transit (P = 9.12438 d)





TIC 154840461 | Sectors: 19, 20 | Phased MCMC Fit - Planet 1

